

Car Price Description

Background

Geely Auto, a prominent Chinese automobile company, aims to expand its operations by entering the U.S. automobile market.

To ensure a successful market entry, the company plans to establish a local manufacturing unit that can produce vehicles competitively with American and European brands.

Before launching, Geely seeks to understand the factors that influence car prices in the U.S. market, as these may differ

significantly from those in China. The company has partnered with an automobile

consulting firm to analyze a dataset

containing various car attributes and their corresponding market prices. This dataset

represents a diverse collection of vehicles from across the American automotive industry.

Business Objective

The primary objective is to develop a predictive model that can accurately estimate the price of a car based on its

attributes. This model will provide strategic insights for Geely's management to understand pricing dynamics and design data-driven business strategies.

By identifying the most influential variables affecting car prices, Geely can make informed decisions about product design,

feature prioritization, and competitive pricing strategies. This analysis also helps the company determine how different features contribute to market value and profitability.

Problem Statement

The dataset includes various independent variables (e.g., engine size, horsepower, fuel type, car body style) and one dependent variable (**price**). Using these variables,

the goal is to:

1. Identify the key factors that significantly influence car prices.
2. Build a **linear regression** model to predict car prices based on features.

Analytical Goals

In this exercise:

- Perform **Exploratory Data Analysis (EDA)** to gain insights into the dataset's structure and relationships.
- Apply **feature engineering and selection** to choose meaningful predictors for the regression model.
- Build, train, and evaluate a **Linear Regression model** to predict car prices.